SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO4	**Assessment:**	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	**Total Marks:**	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	A. Govardhan Reddy	**Reg. No.:**	192421359
Date:	13 – 08 - 2026	**Duration:**	60 minutes

Code of Conduct

I A. Govardhan Reddy (192421359) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A. Govardhan Reddy

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

1. Problem Statement

Airport runway and ground operations currently depend on fragmented systems and manual coordination between air traffic control, ground crews, and vehicle operators, increasing the risk of runway incursions and operational delays. Existing surveillance systems lack real-time integration between aircraft position data, ground vehicle movement, and weather conditions, and offer minimal predictive capability, causing reactive rather than preventive safety management. Ground delays are not systematically correlated with runway scheduling, and incident data is recorded manually, slowing trend analysis. A unified platform is needed to consolidate runway safety monitoring, ground operations tracking, and predictive risk alerts into a single real-time system.

2. Requirements Identified for Testing

ID	Requirement Description
FR1	Track real-time aircraft position on runway/taxiway
FR2	Track real-time ground vehicle movement (tugs, fuel trucks, baggage carts)
FR3	Integrate live weather data into hazard detection logic
FR4	Generate predictive conflict / runway-incursion alerts
FR5	Correlate ground-ops events (pushback, fueling) with runway scheduling
FR6	Automatically log incidents and near-misses
FR7	Provide a unified dashboard for ATC, ground crew, and vehicle operators
FR8	Deliver alert notifications to the correct personnel/role
NFR1	Real-time performance — alert generated within 2 seconds of hazard detection
NFR2	High availability / reliability — 24x7 uptime, graceful handling of sensor loss
NFR3	Scalability — supports peak-traffic volumes of aircraft and vehicles
NFR4	Data accuracy — position and weather data synchronized within tolerance
NFR5	Usability — dashboard reduces controller cognitive load
NFR6	Security — role-based access control for safety-critical data
NFR7	Auditability — historical incident data retrievable for trend analysis

3. Test Scenarios

- TS1: Detect and alert potential runway incursion — aircraft vs aircraft
- TS2: Detect and alert potential incursion — aircraft vs ground vehicle
- TS3: Weather condition (low visibility) triggers hazard alert
- TS4: Weather condition (high wind speed) triggers hazard alert
- TS5: System tracks multiple simultaneous assets without missing a conflict
- TS6: Predictive alert fires with sufficient lead time before conflict escalates
- TS7: Late pushback/fueling is correlated with runway schedule to flag cascading delay
- TS8: Incident/near-miss is auto-logged and retrievable for trend analysis
- TS9: System handles invalid/missing sensor (position) data feed
- TS10: Behaviour at minimum safe separation distance (boundary)
- TS11: System performance under peak traffic load
- TS12: Unauthorized user attempts dashboard access
- TS13: Alert notification is routed to the correct role
- TS14: Aircraft ground-status transitions are tracked correctly

4. Detailed Test Cases

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC01	Aircraft-aircraft incursion (normal)	1. Two aircraft on intersecting runway paths 2. System	AC1 pos: RWY09 hold AC2 pos:	System raises incursion alert to ATC within 2s	Alert raised	Pass

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
		computes separation 3. Separation falls below safe threshold	RWY09 approach, sep=250m (threshold=300m)			
TC02	Aircraft-vehicle incursion (normal)	1. Fuel truck crosses active runway 2. Aircraft on final approach 3. System evaluates positions	Vehicle on RWY27, Aircraft 1200m out on approach	Incursion alert raised to ATC and vehicle operator	Alert raised	Pass
TC03	Low-visibility weather alert (normal)	1. Fetch live visibility reading 2. Compare to safe-ops threshold 3. Trigger hazard flag if below limit	Visibility = 400m (threshold = 550m)	Hazard alert generated; low-visibility procedures flagged	Alert raised	Pass
TC04	High-wind weather alert (normal)	1. Fetch live wind speed 2. Compare to safe-ops threshold 3. Trigger hazard flag if exceeded	Wind speed = 45 knots (threshold = 35 knots)	Hazard alert generated for ground ops and ATC	Alert raised	Pass
TC05	Multi-asset simultaneous tracking (normal/load)	1. Load 50 simultaneous aircraft/vehicle feeds 2. Introduce one genuine conflict 3. Verify system still detects it	50 active asset feeds, 1 conflicting pair	Conflict detected and alerted; no missed detections	Conflict detected	Pass
TC06	Predictive alert lead time (normal)	1. Simulate converging trajectories 2. Measure time between alert and projected conflict	Closing speed = 20 knots, projected conflict in 40s	Alert issued ≥ 20s before projected conflict	Alert at T-25s	Pass
TC07	Ground delay to runway-schedule correlation (normal)	1. Log pushback start time later than scheduled 2. System checks impact on runway slot	Scheduled pushback 10:00, actual 10:12 (12 min late)	System flags cascading delay risk to runway schedule	Delay flagged	Pass
TC08	Incident/near-miss auto-log (normal)	1. Trigger a near-miss event 2. Query incident log/database	Near-miss event ID NM-1042	Event auto-logged with timestamp, location, assets involved; retrievable via query	Logged & retrieved	Pass
TC09	Sensor/position feed drop (exceptional)	1. Simulate loss of aircraft position feed 2. Observe system response	Aircraft feed interrupted for 10s	System flags stale/lost data, alerts controller, does not silently fail	Stale-data warning shown	Pass
TC10	Separation at exact threshold (boundary - BVA)	1. Set separation exactly at safe minimum 2. Observe alert behaviour	Separation = 300m (threshold = 300m, min-1=299, min+1=301)	System treats as boundary-safe per spec (no false alert at exact minimum)	No alert	Pass
TC11	Separation just below minimum (boundary - BVA)	1. Set separation to threshold-1 2. Observe alert behaviour	Separation = 299m	Incursion alert raised	Alert raised	Pass
TC12	Separation just above minimum (boundary - BVA)	1. Set separation to threshold+1 2. Observe alert behaviour	Separation = 301m	No alert raised	No alert	Pass
TC13	Peak traffic load performance (exceptional/stress)	1. Simulate peak-hour asset count 2. Measure alert latency and system responsiveness	200 simultaneous tracked assets	Alert latency remains ≤ 2s; no dropped updates	Latency 1.8s	Pass
TC14	Unauthorized dashboard access (invalid/security)	1. Attempt login with invalid/unauthorized credentials 2. Attempt to view restricted safety data	Invalid credentials / unauthorized role	Access denied; attempt logged for audit	Access denied	Pass
TC15	Alert routed to correct role (normal - decision table)	1. Generate alerts for runway, vehicle, and weather hazard types 2. Verify recipient per hazard/zone combination	Hazard type=Runway incursion, Zone=Active runway	Alert routed to ATC + affected vehicle operator only (per decision table)	Routed correctly	Pass
TC16	Aircraft ground-status transition (state transition)	1. Move aircraft through Parked→Taxi→Runway Entry→Takeoff 2. Attempt invalid transition Parked→Takeoff directly	Status sequence input incl. one invalid jump	Valid transitions accepted & tracked; invalid transition rejected/flagged	Invalid transition flagged	Pass

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC17	Invalid weather data values (invalid - EP)	1. Submit out-of-range weather reading 2. Observe system validation	Wind speed = -15 knots (invalid)	System rejects/flags invalid input; does not use it for hazard calc	Input rejected	Pass
TC18	Conflicting ground-vehicle position feed (exceptional)	1. Two feeds report different positions for same vehicle ID 2. Observe system handling	Vehicle V-12: Feed A=RWY09, Feed B=Taxiway C	System flags data conflict for review; does not silently pick one	Conflict flagged	Pass

5. Test Design Techniques Applied

- Equivalence Partitioning — weather parameters (visibility, wind speed) partitioned into valid/invalid classes; representative values tested in TC03, TC04, TC17.
- Boundary Value Analysis — minimum safe separation distance tested at threshold, threshold-1, and threshold+1 in TC10, TC11, TC12.
- Decision Table Testing — alert routing logic (hazard type × zone → recipient) validated in TC15.
- State Transition Testing — aircraft ground-status lifecycle (Parked → Taxi → Runway Entry → Takeoff) and invalid transitions validated in TC16.

6. Traceability Matrix

Requirement ID	Covered By Test Case(s)
FR1	TC01, TC02, TC05, TC09, TC16
FR2	TC02, TC05, TC18
FR3	TC03, TC04, TC17
FR4	TC01, TC02, TC03, TC04, TC06, TC10, TC11, TC12, TC16
FR5	TC07
FR6	TC08
FR7	TC05 (dashboard used across scenarios)
FR8	TC15
NFR1	TC01, TC06, TC13
NFR2	TC09
NFR3	TC05, TC13
NFR4	TC10, TC18
NFR5	TC05, TC15 (usability observed operationally)
NFR6	TC14
NFR7	TC08

7. Test Coverage Summary

Total Requirements (FR + NFR)	15
Requirements Covered	15 (100%)
Total Test Cases Designed	18
Normal Condition Cases	8 (TC01–TC08)
Boundary Condition Cases	3 (TC10–TC12)
Invalid / Exceptional Condition Cases	7 (TC09, TC13, TC14, TC17, TC18 + TC15/TC16 partial)